IDEATION PHASE

Empathize & Discover — Empathy Map

DATE	12 July 2026
TEAM ID	
PROJECT NAME	**Plugging into the Future: An Exploration of Electricity Consumption Patterns Using Tableau**
MAXIMUM MARKS	4 Marks

Empathy Mapping Summary:
The Empathy Map was created to understand the needs and challenges of energy analysts and planners while analyzing electricity consumption data. It helped identify key pain points such as difficulty in interpreting raw data and lack of proper visualization tools. This understanding guided the development of an interactive Tableau dashboard that provides clear and meaningful insights.

TARGET USER: ENERGY ANALYST / GOVERNMENT ENERGY PLANNER

WHAT THEY THINK

- Needs clear understanding of electricity consumption trends.
- Wants accurate comparison between 2019 and 2020.
- Wonders which region consumes the most electricity.

WHAT THEY SEE

- Large Excel sheets filled with raw usage data metrics.
- Different states/regions showing highly varied consumption.
- No clear interactive dashboard visualization to compare trends easily.
- Reports that are complex and hard to interpret quickly.

WHAT THEY HEAR

- Government emphasis on grid efficiency and energy optimization.
- Discussions surrounding localized power shortages.
- Suggestions about renewable energy source adoption paradigms.
- Pressure to enhance baseline energy distribution frameworks.

WHAT THEY DO

- Collects and reviews regional consumption datasets.
- Uses standard Microsoft Excel layouts for preliminary data cleanup.
- Compares year-over-year operational trends via manual operations.
- Formulates tactical planning choices relying on restricted static insight lookups.

⚠ PAINS

- Difficult and laborious to parse expansive industrial text sheets manually.

⊛ GAINS

- Interactive, highly visual Tableau analytical workspace.

• Absence of centralized dashboards to quickly discover operational patterns. • Time-consuming data cross-referencing routines across mixed sectors. • Hard to filter regional demand fluctuations on the fly.	• Clear side-by-side year-wise (2019 vs 2020) and macro-region evaluation. • Rapid parameter-driven filtering of Top N and Bottom N data slices. • Accessible monthly and seasonal trend visualization.